Non-Gaussian critical-point statistics for parameter inference

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Filaments in multi-scale Universe,

@ Kavli IPMU, Oct. 30 2026

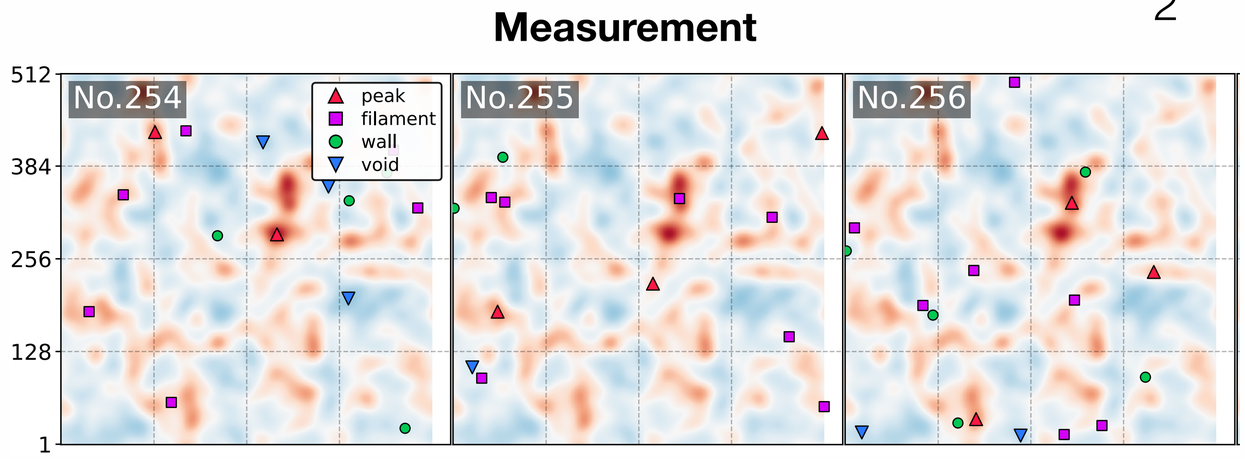
Based in part on Y. He and T. Matsubara,
"Validating weakly non-Gaussian critical-point statistics in cosmic fields",
submitted to Physical Review D (2026).

1st-order critical-point statistics

Critical point: $\nabla f = 0$,

classified by $H = \nabla\nabla f$ eigenvalues' sign

for $\alpha = f/\sigma_0 > \nu$, the abundance is



$$\bar{n}^{\star}(\nu) = \left\langle \Theta(\alpha-\nu)\,\delta_{\rm D}(\nabla f)\,\Theta(\lambda^{\star})\left|\det H\right|\right\rangle$$

BBKS (1986), Adler & Taylor (2007), Pogosyan et al. (2009), Gay et al. (2012), Matsubara (2020)

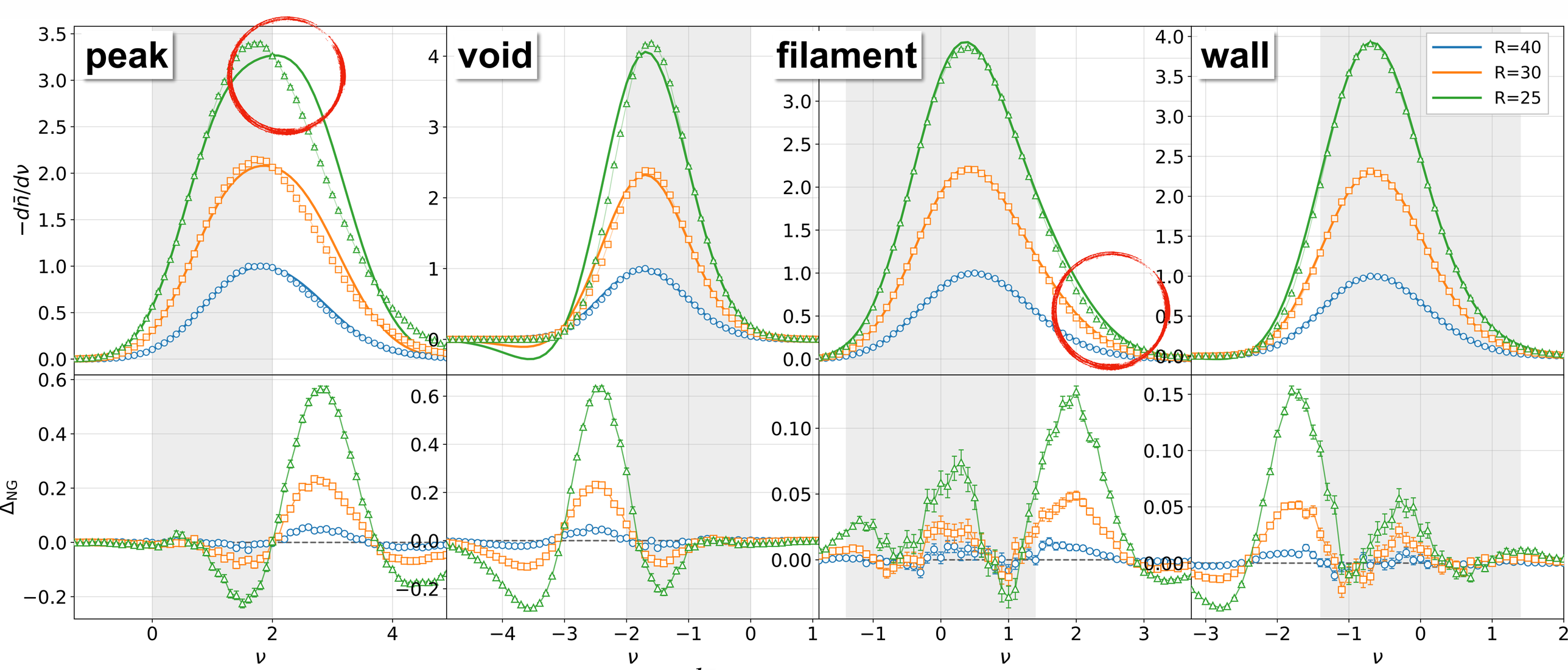
1. Construct rotational invariants
2. Expend joint PDF
   using Gram-Charlier expansion
3. Integrate over invariants

$$\bar{n}^{\star}(\nu) = G^{\star}_{00000} + \frac{\sigma_0}{6}\sum_a S^{(a)} G^{\star}_a + \mathcal{O}(\sigma_0{}^2)$$

Gaussian prediction

generalized skewness parameters

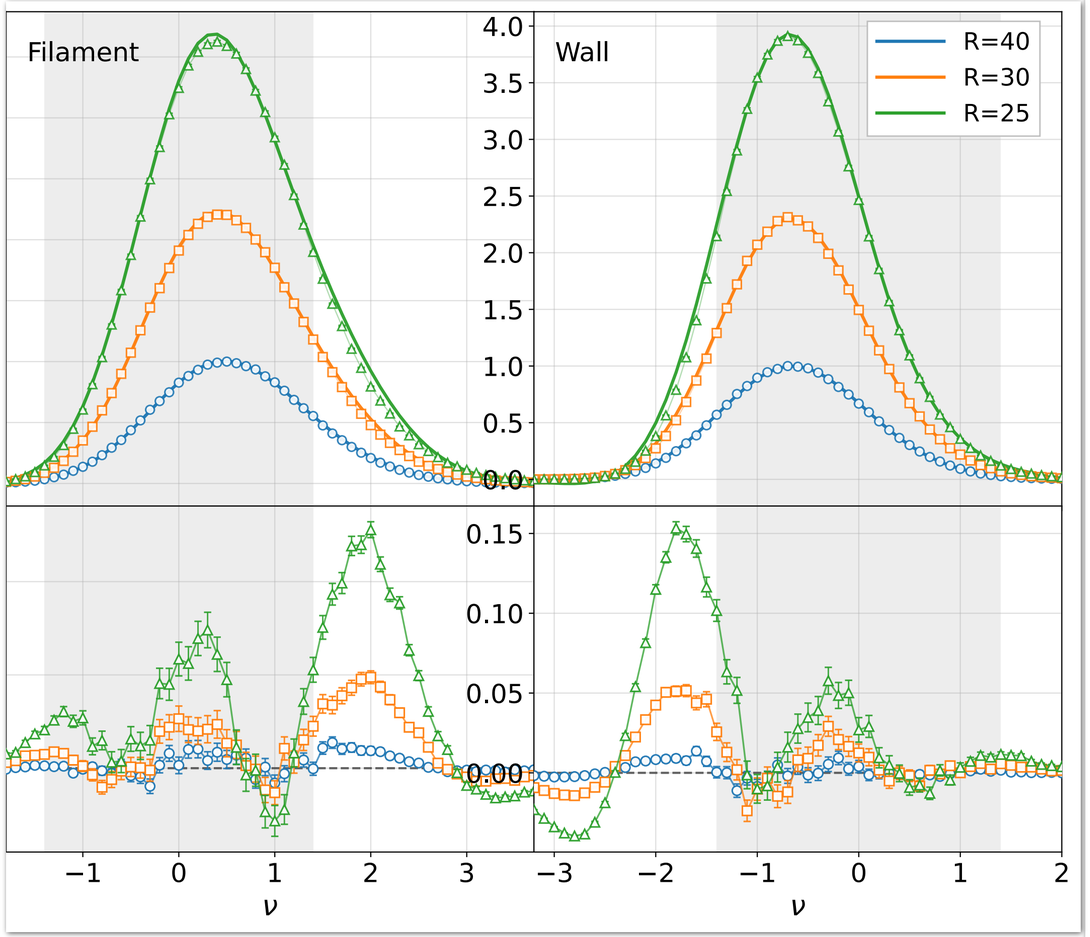
Gaussian response coefficients



Differential number density $-\dfrac{d\bar{n}}{d\nu}$

Using validated region (shaded)
for parameter inference

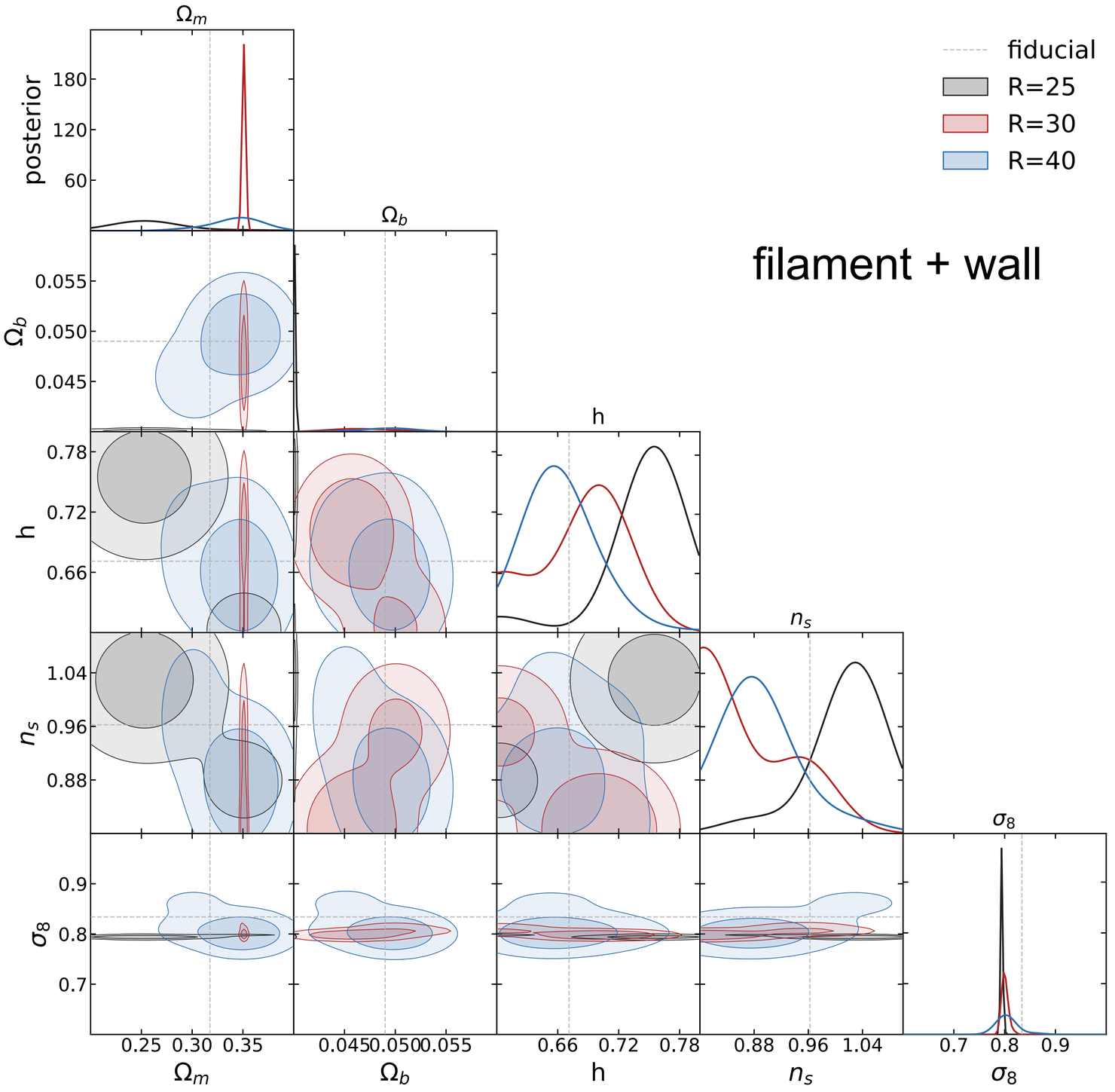
R = 20, 30, 40, σ_0 = 0.15, 0.12, 0.08
N_sim = 2000, L_box = 1000, N_grid = 512

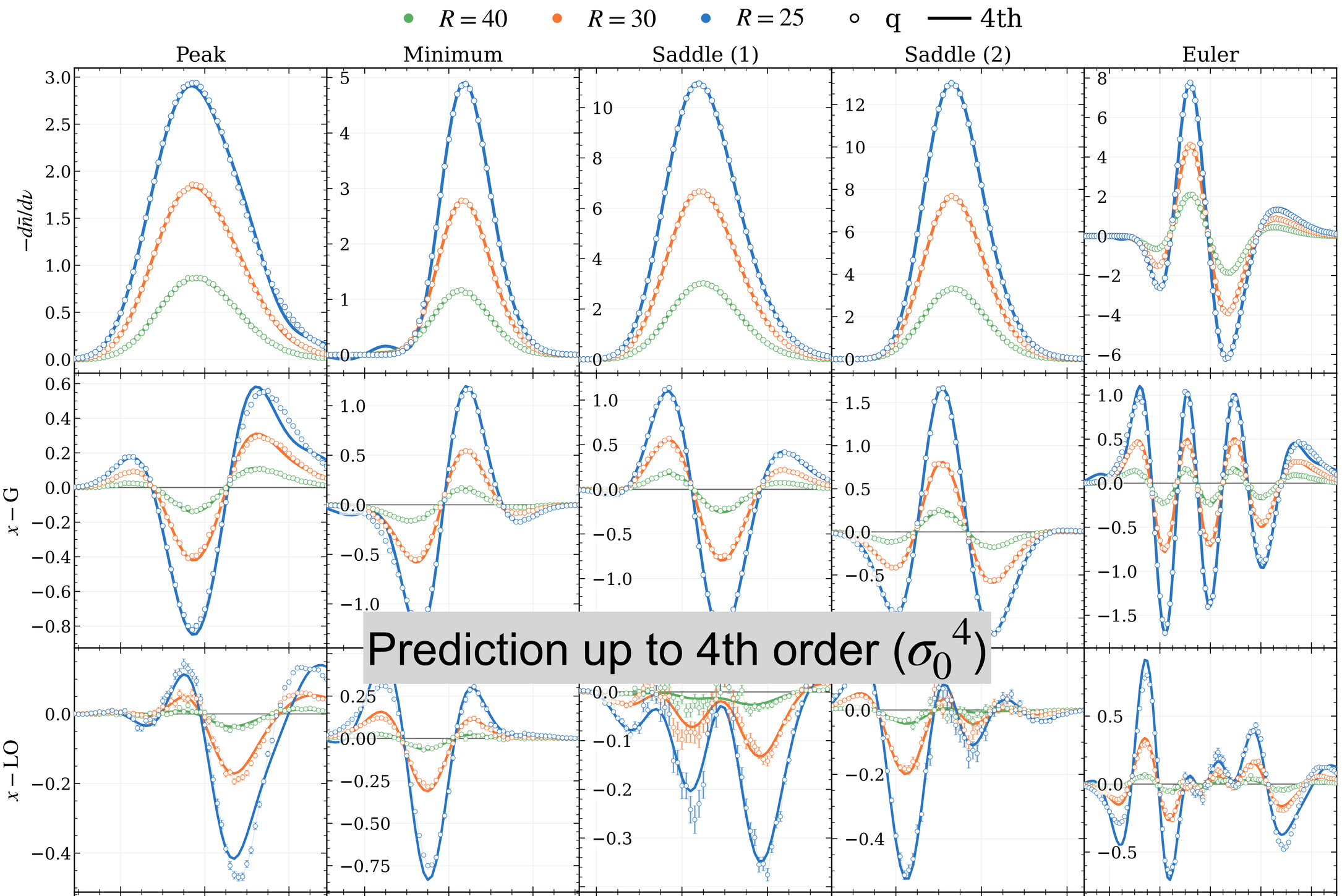


Smaller R, higher constraints,

but larger biases

→ Higher-order corrections are needed




$R = 40$
$R = 30$
$R = 25$
q
4th
Peak
Minimum
Saddle (1)
Saddle (2)
Euler
$-d\bar{n}/d\nu$
$x - \mathrm{G}$
$x - \mathrm{LO}$
Prediction up to 4th order ($\sigma_0{}^4$)

Conclusion

- Leading-order predictions become less accurate for peaks and voids in mildly non-Gaussian regime, while filaments and walls remain robust

- Within the validated regime, combined critical-point statistics constraint $(\Omega_{\mathrm{m}}, \Omega_{\mathrm{b}}, h, n_s, \sigma_8)$ and helps reduce parameter degeneracies

- We are extending the calculation up to fourth order, with AI-assisted derivations and validation

Thank you :)